

```
In [1]: list1 = [] # empty list
```

```
In [2]: list2 = [10,20,30,40,50] # list of int
```

```
In [3]: list3 = [1.1,2.2,3.3,4.4,5.5] # list of float
```

```
In [5]: list4 = ['one','two','three'] # list of str
```

```
In [6]: list5 = [10,20,30,[1,2,3]] # nextedlist
```

```
In [7]: list6 = [10,1.3,'one'] # list of mixed data types
```

list indexing

```
In [9]: list2[1]
```

```
Out[9]: 20
```

```
In [10]: list3[-1]
```

```
Out[10]: 5.5
```

list slicing

```
In [13]: mylist = ['one' , 'two' , 'three' , 'four' , 'five' , 'six' , 'seven']
```

```
In [14]: mylist[0:3]
```

```
Out[14]: ['one', 'two', 'three']
```

```
In [15]: mylist[-1]
```

```
Out[15]: 'eight'
```

```
In [16]: mylist[-3:]
```

```
Out[16]: ['six', 'seven', 'eight']
```

```
In [17]: mylist[::-1]
```

```
Out[17]: ['eight', 'seven', 'six', 'five', 'four', 'three', 'two', 'one']
```

Add , Remove & Change Items

```
In [18]: mylist
```

```
Out[18]: ['one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight']
```

```
In [19]: mylist.append('nine')
```

```
In [20]: mylist
```

```
Out[20]: ['one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight', 'nine']
```

```
In [21]: mylist.insert(9, 'ten')
mylist
```

```
Out[21]: ['one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight', 'nine', 'ten']
```

```
In [22]: mylist.insert(1, 'ONE')
mylist
```

```
Out[22]: ['one',
          'ONE',
          'two',
          'three',
          'four',
          'five',
          'six',
          'seven',
          'eight',
          'nine',
          'ten']
```

```
In [23]: mylist.remove('ONE')
```

```
In [24]: mylist
```

```
Out[24]: ['one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight', 'nine', 'ten']
```

```
In [25]: mylist.pop()
mylist
```

```
Out[25]: ['one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight', 'nine']
```

```
In [26]: mylist.pop(8)
mylist
```

```
Out[26]: ['one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight']
```

```
In [27]: del mylist[7]
```

```
In [28]: mylist
```

```
Out[28]: ['one', 'two', 'three', 'four', 'five', 'six', 'seven']
```

```
In [29]: # change the values
mylist[0] = 0
mylist[1] = 1
mylist[2] = 2
mylist
```

```
Out[29]: [0, 1, 2, 'four', 'five', 'six', 'seven']
```

```
In [30]: mylist.clear()  
mylist
```

```
Out[30]: []
```

```
In [31]: del mylist
```

```
In [32]: mylist
```

```
-----  
NameError                                Traceback (most recent call  
l last)  
Cell In[32], line 1  
----> 1 mylist  
  
NameError: name 'mylist' is not defined
```

Copy list

```
In [34]: mylist = ['one', 'two', 'three', 'four', 'five', 'six', 'seven', 'e
```

```
In [36]: mylist1 = mylist # Create a new reference using mylist1
```

```
In [38]: id(mylist) , id(mylist1) # both address will be same
```

```
Out[38]: (5597035584, 5597035584)
```

```
In [39]: mylist2 = mylist.copy()
```

```
In [40]: id(mylist2) # it will different because copy function
```

```
Out[40]: 5597099264
```

Join Lists

```
In [41]: list1 = ['one', 'two', 'three', 'four']  
list2 = ['five', 'six', 'seven', 'eight']
```

```
In [42]: list1.extend(list2)
```

```
In [43]: list1
```

```
Out[43]: ['one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight']
```

```
In [44]: list3 = list1 + list2
```

```
In [45]: list3
```

```
Out[45]: ['one',  
          'two',  
          'three',  
          'four',  
          'five',  
          'six',  
          'seven',  
          'eight',  
          'five',  
          'six',  
          'seven',  
          'eight']
```

List Membership

```
In [46]: list1
```

```
Out[46]: ['one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight']
```

```
In [47]: 'one' in list1
```

```
Out[47]: True
```

```
In [48]: 'nine' in list1
```

```
Out[48]: False
```

```
In [49]: if 'one' in list1:  
          print('one is present in the list')  
else:  
    print('in these not there')
```

one is present in the list

Reverse & Sort List

```
In [50]: mylist
```

```
Out[50]: ['one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight', 'nine']
```

```
In [54]: mylist.reverse()  
mylist
```

```
Out[54]: ['eight', 'five', 'four', 'nine', 'one', 'seven', 'six', 'three', 'two']
```

```
In [55]: mylist[::-1]
```

```
Out[55]: ['two', 'three', 'six', 'seven', 'one', 'nine', 'four', 'five', 'eight']
```

loop through list

```
In [56]: list1
```

```
Out[56]: ['one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight']
```

```
In [57]: for i in list1:  
         print(i)
```

```
one  
two  
three  
four  
five  
six  
seven  
eight
```

```
In [58]: for i in enumerate(list1):  
         print(i)
```

```
(0, 'one')  
(1, 'two')  
(2, 'three')  
(3, 'four')  
(4, 'five')  
(5, 'six')  
(6, 'seven')  
(7, 'eight')
```

```
count
```

```
In [59]: list10=['one', 'two', 'three', 'four', 'one', 'one', 'two', 'three
```

```
In [60]: list10.count('one')
```

```
Out[60]: 3
```

```
In [ ]:
```